SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1021	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	28-10-26	**Duration:**	60 minutes

Code of Conduct

I S.Venkateswarlu (1924253402)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___S.Venkateswarlu_____

Annexure A – Scenario-Based Requirement Analysis

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.

Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

Industry Context: International seaports are highly complex bottlenecks where giant container ships must be loaded and unloaded quickly. Inefficient container yard management leads to crane delays, traffic jams for terminal trucks, and long ship wait times, costing shipping companies thousands of dollars per hour. Port operators need a centralized, automated orchestration system that schedules crane movements and container placements to maximize port speed.

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Problem Statement: Design and develop a highly responsive Seaport Container Yard Coordination Platform that orchestrates container movements, crane schedules, and cargo truck traffic. The system must process real-time inputs—including ship cargo configurations, automated guided vehicle (AGV) paths, container yard storage grids, and import/export timelines. It must compute optimal crane picking patterns to minimize ship unloading times.

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Objectives:

o
Balance
container placements inside yard storage grids to minimize redundant movements.

o
Coordinate
crane movements in real time to avoid collisions and mechanical downtime.

o
Route
automated terminal trucks through cargo yards using dynamic traffic paths.

o
Adapt
yard logistics automatically when ship arrivals shift due to weather delays.

o

Provide
live operational dashboards for port logistics and customs management teams.

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Expected Outcomes:

o
Shorter
turnaround times for cargo ships at international maritime shipping ports.

o
Maximum
performance and utilization from high-value port cranes and automated trucks.

o
Lower
traffic gridlock and fuel waste for terminal transport vehicles inside ports.

o
Increased
cargo processing capacities for ports without expanding physical footprints.

Software Requirements Specification (SRS)

Seaport Container Yard Coordination Platform

Project Type: Scenario-Based Requirement Analysis

Domain: Maritime Logistics, Port Operations, Container Terminal Management, Real-Time Scheduling and Optimization

1. Study and Analyze the Scenario

1.1 Introduction

International seaports are complex bottlenecks where giant container ships must be loaded and unloaded quickly. Inefficient yard management causes crane delays, traffic jams for terminal trucks and long vessel waiting times, costing shipping companies thousands of dollars per hour. Container placement, crane sequencing and truck routing are normally planned separately, so a delay in one area spreads to the others. The Seaport Container Yard Coordination Platform is a centralized, real-time orchestration system. It takes in ship cargo configurations (bay plans), AGV/terminal-truck paths, yard storage grid states and import/export timelines. From these it computes container slot allocations, crane picking sequences and truck routes, re-plans automatically when vessel arrivals shift (for example because of weather), and shows live dashboards to port logistics and customs teams.

1.2 Business Domain

The system belongs to the following domains:

- Maritime port and container terminal operations
- Container yard storage and inventory management
- Quay crane, yard crane and gantry scheduling
- Automated guided vehicle (AGV) and terminal truck fleet management
- Customs clearance and cargo release management
- Real-time operational analytics and decision support

1.3 Stakeholders

Stakeholder	**Role / Interest**
Port terminal operator / Port Operations Manager	Owns throughput targets; wants shorter vessel turnaround and higher crane and truck utilization.
Shipping lines and vessel agents	Supply cargo manifests and bay plans; want minimal ship waiting and berth time.
Crane operators and controllers	Execute crane moves; need safe, conflict-free sequences.
Yard planners	Manage storage grids and container stacking policies.
Terminal truck / AGV fleet	Automated vehicles that need dynamic, congestion-free routes.
Customs and border authorities	Place holds, inspect and release containers; need live container status.
Weather and port authority services	Provide weather, berth and arrival-time updates.
System administrators	Maintain configuration, users, integrations and availability.
Safety, compliance and audit officers	Review incidents, decisions and regulatory records.

1.4 Existing Challenges

- Containers placed without considering their departure order cause redundant re-handling ("shuffling") moves.
- Cranes work without a shared real-time view, leading to idle time, collisions and conflicts on shared rails.
- Terminal trucks and AGVs queue at crane and yard blocks, creating gridlock and wasting fuel.
- Plans are static; a delayed or early vessel invalidates berth, crane and yard allocations.
- Customs holds are communicated late, so held containers block the yard.
- Operations staff and customs have no single live view of yard, crane and fleet status.

- Physical yard space is fixed, so capacity must come from better use of what exists.

1.5 System Objectives

- Balance container placements in yard storage grids to minimize redundant movements.
- Coordinate crane movements in real time to avoid collisions and mechanical downtime.
- Route automated terminal trucks through the yard using dynamic traffic paths.
- Adapt yard logistics automatically when ship arrivals shift because of weather delays.
- Provide live operational dashboards for port logistics and customs management teams.
- Compute crane picking patterns that minimize ship unloading time.

Expected outcomes: shorter ship turnaround, higher utilization of cranes and automated trucks, less traffic gridlock and fuel waste, and higher cargo processing capacity without expanding the physical footprint. Targets must be confirmed through simulation and pilot testing.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

ID	Requirement
FR-01	The system shall ingest ship cargo configurations (bay plans, container lists, discharge/load sequences).
FR-02	The system shall ingest vessel schedules (ETA/ETD), berth availability and import/export timelines.
FR-03	The system shall receive real-time yard grid state, crane positions and AGV/truck locations and paths.
FR-04	The system shall validate and timestamp incoming data and flag missing, stale or conflicting records.
FR-05	The system shall maintain a digital model of yard blocks, bays, rows and tiers with container positions.

FR-06	The system shall recommend a yard slot for each container using departure time, weight, type, destination and customs status.
FR-07	The system shall balance load across yard blocks and minimize expected re-handling (shuffling) moves.
FR-08	The system shall compute crane picking sequences that minimize ship unloading and loading time.
FR-09	The system shall schedule crane moves in real time and detect potential crane-to-crane and crane-to-vehicle conflicts.
FR-10	The system shall avoid collisions by re-sequencing, delaying or reassigning crane tasks and alerting operators.
FR-11	The system shall take cranes with faults or maintenance windows out of the schedule and reassign their tasks.
FR-12	The system shall assign transport jobs to AGVs/terminal trucks and compute dynamic routes through the yard.
FR-13	The system shall re-route vehicles in real time around congestion, blockages and closed lanes.
FR-14	The system shall detect vessel delay or early arrival (including weather-related changes) and trigger automatic re-planning.
FR-15	The system shall re-allocate berths, crane assignments, yard slots and truck jobs after a schedule change and publish the revised plan.
FR-16	The system shall record customs holds, inspections and releases and exclude held containers from pick plans.
FR-17	The system shall provide live dashboards for ship progress, crane and fleet utilization, yard occupancy and alerts.
FR-18	The system shall provide a customs view showing container status, location and hold/release state.
FR-19	The system shall notify authorized users of delays, conflicts, equipment faults and safety alerts.
FR-20	The system shall allow authorized planners to review, override or approve generated plans, and shall log overrides.

FR-21	The system shall log all plans, decisions, vehicle and crane commands for audit and performance reporting.
FR-22	The system shall support role-based access control and configurable operating rules (stacking policy, speed zones, safety clearances).

2.2 Non-Functional Requirements

ID	Requirement	Target or acceptance criterion
NFR-01	Performance / responsiveness	Dashboards refresh within 2 seconds of an event; crane/vehicle re-planning proposals within a defined bound (e.g., 5 seconds) under benchmark load.
NFR-02	Scalability	Support multiple berths, dozens of cranes and hundreds of AGVs/trucks and tens of thousands of containers by adding compute nodes.
NFR-03	Availability	24x7 operation with a defined availability target (e.g., 99.9%) and recovery procedure.
NFR-04	Reliability / fault tolerance	Continue safe operation on sensor loss, network drops or node failure; fall back to last valid plan and manual control.
NFR-05	Safety	Collision-avoidance rules take priority over throughput; commands are rejected if they violate safety clearances.
NFR-06	Security	Authenticate users and devices, apply least privilege, encrypt data in transit and at rest, and protect against tampering with control commands.
NFR-07	Data integrity and consistency	All planning uses a time-stamped consistent snapshot; container IDs and positions are validated against the yard model.
NFR-08	Usability	Role-specific dashboards that operators can use with minimal training; clear colour-coded alerts.
NFR-09	Interoperability	Integrate with terminal operating system, crane PLCs, AGV fleet controllers, customs systems and weather feeds through documented APIs.

NFR-10	Maintainability	Modular components, versioned configuration, automated tests and documented APIs.
NFR-11	Auditability	Retain decisions, overrides and commands according to port and customs retention policy.
NFR-12	Efficiency	Optimization should reduce measurable re-handling moves, vehicle idle time and fuel use compared with the baseline.

3. Actors and Use Cases

3.1 Primary and Secondary Actors

Actor	Type	Responsibilities
Port Operations Manager	Primary	Monitors performance, approves re-plans, configures operating rules.
Yard Planner	Primary	Reviews slot allocations and stacking policies; adjusts plans.
Crane Operator / Controller	Primary	Executes crane tasks and responds to conflict alerts.
Customs Officer	Primary	Places holds, tracks container status and releases cargo.
System Administrator	Primary	Manages users, devices, integrations and configuration.
Shipping Line / Vessel Agent	Secondary	Supplies manifests, bay plans and schedule changes.
Terminal Truck / AGV Fleet Controller	Secondary	Receives jobs and routes; reports position and status.
Weather Service	Secondary	Provides forecasts and alerts that affect vessel arrival.
Auditor	Secondary	Inspects logs, decisions and incident records.

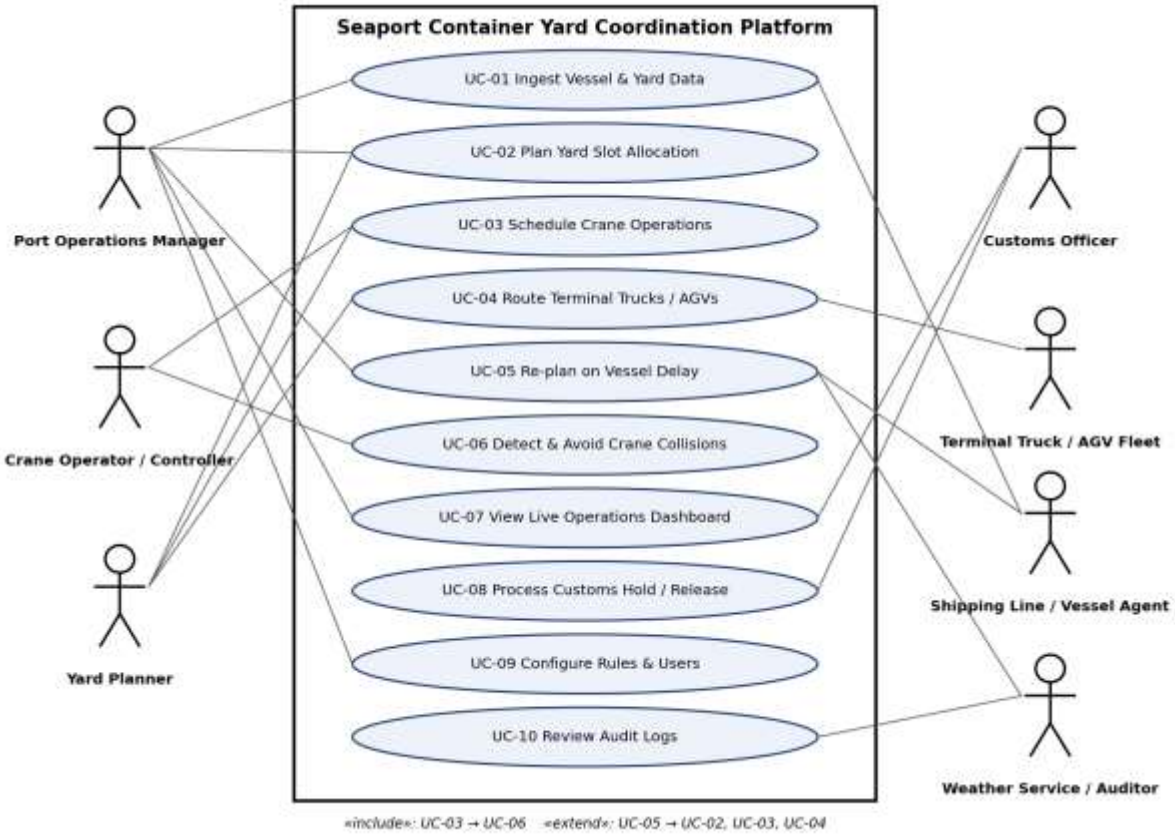
3.2 Major Use Cases

Use Case ID	Use Case	Actor
UC-01	Ingest vessel, cargo and yard data	Shipping Line, Port Operations Manager
UC-02	Plan yard slot allocation	Yard Planner, Port Operations Manager
UC-03	Schedule crane operations	Crane Operator, Yard Planner
UC-04	Route terminal trucks / AGVs	AGV Fleet Controller, Yard Planner

UC-05	Re-plan on vessel delay	Weather Service, Shipping Line, Port Operations Manager
UC-06	Detect and avoid crane collisions	Crane Operator
UC-07	View live operations dashboard	Port Operations Manager, Customs Officer
UC-08	Process customs hold / release	Customs Officer
UC-09	Configure rules and users	Port Operations Manager, System Administrator
UC-10	Review audit logs	Auditor

3.3 Use Case Diagram

The following diagram shows the principal interactions between users, external systems and the platform.



3.4 Use Case Description: Schedule Crane Operations

Field	**Description**

Use Case ID	UC-03
Use Case Name	Schedule Crane Operations
Primary Actor	Crane Operator / Controller
Supporting Actors	Yard Planner, Shipping Line, AGV Fleet Controller
Precondition	Vessel bay plan, yard state, crane availability and AGV status are available and valid.
Trigger	A vessel berths, a crane becomes available or unavailable, or the plan changes.
Main Flow	1. Load bay plan and discharge/load sequence. 2. Read crane positions, status and safety zones. 3. Compute a picking sequence that minimizes unloading time. 4. Check crane-to-crane and crane-to-vehicle conflicts. 5. Match crane tasks with AGV pickup times and yard slots. 6. Publish the schedule to operators and the dashboard. 7. Monitor progress and log results.
Alternative Flow	If a conflict or fault is detected, re-sequence or reassign tasks, alert the operator and mark the revised plan. If data is stale, use the last valid plan and request manual confirmation.
Postcondition	A conflict-free, time-stamped crane schedule is active and recorded.

3.5 Use Case Description: Re-plan on Vessel Delay

Field	**Description**
Use Case ID	UC-05
Use Case Name	Re-plan on Vessel Delay
Primary Actor	Port Operations Manager
Supporting Actors	Weather Service, Shipping Line, Yard Planner
Precondition	A baseline berth, crane, yard and truck plan exists.
Trigger	A weather alert or updated ETA shifts a vessel arrival beyond a configured threshold.
Main Flow	1. Receive the delay or early-arrival notice. 2. Validate the new ETA. 3. Recalculate berth windows and crane assignments. 4. Adjust yard slot reservations and import/export pick-up windows. 5. Reassign truck and

	AGV jobs and routes. 6. Present the revised plan for approval or auto-apply within configured limits. 7. Notify affected users and update dashboards.
Alternative Flow	If no feasible plan meets constraints, propose the best alternative, flag affected vessels and require manager decision.
Postcondition	A revised, approved plan is active and the change is audited.

4. Software Requirement Specification (SRS)

4.1 Introduction

The purpose of this SRS is to define the functional capabilities, quality requirements, interfaces, data and operating constraints of the Seaport Container Yard Coordination Platform. It is a reference for design, implementation, testing and acceptance.

4.2 Problem Description

Ship turnaround depends on how well container placement, crane work and truck movement are synchronized. Poor slot choices cause extra moves, cranes can block each other or stand idle, and trucks queue in congested lanes. Late or early vessels invalidate fixed plans. Port operators need a centralized and automated coordination system that plans and adapts continuously to maximize speed and equipment use.

4.3 Scope of the System

Included in scope:

- Ingestion of cargo, vessel, yard, crane and vehicle data.
- Yard slot allocation and re-handling minimization.
- Crane sequencing, conflict detection and collision avoidance.
- Dynamic routing of terminal trucks and AGVs.
- Automatic re-planning for vessel schedule changes.
- Live dashboards for port logistics and customs teams.

- Audit logging and administration.

Outside the initial scope:

- Physical construction or expansion of port infrastructure.
- Replacing low-level crane PLC or vehicle safety controllers.
- Vessel navigation, tugging and pilotage.
- Customs clearance decisions (the system only records and shows status).
- Guaranteeing weather forecast accuracy.

4.4 System Features

Feature	**Description**
Data Ingestion and Validation	Collects and validates manifests, schedules, yard and equipment data.
Yard Slot Optimizer	Allocates and balances container positions to reduce re-handling.
Crane Scheduler	Computes picking sequences and real-time crane task schedules.
Collision and Downtime Manager	Detects conflicts and faults and re-sequences or reassigns work.
Truck / AGV Router	Assigns jobs and computes dynamic, congestion-aware routes.
Delay Adaptation Engine	Re-plans berths, cranes, yard and trucks when arrivals shift.
Live Dashboards	Role-based views for operations and customs teams.
Audit and Administration	Logs decisions, manages users, rules and integrations.

4.5 Data Requirements

Data Entity	**Important Attributes**	**Purpose**
Vessel	Vessel ID, ETA/ETD, berth, bay plan, status	Berth and crane planning

Container	Container ID, size/type, weight, origin/destination, import/export flag, departure time, customs status, position	Tracking and slot allocation
Yard Block / Slot	Block, bay, row, tier, occupancy, restrictions	Storage grid model
Crane	Crane ID, type, position, status, safety zone, maintenance window	Crane scheduling
AGV / Truck	Vehicle ID, position, battery/fuel, status, current job	Fleet routing
Route / Path	Segment, direction, speed limit, congestion, closures	Dynamic routing
Task / Job	Job ID, container, source, destination, assigned resource, status, time	Orchestration
Customs Record	Container ID, hold/release, inspection status, timestamp	Customs control
Weather / Schedule Event	Event type, affected vessels, time, severity	Re-planning triggers
Audit Log	Event, user/system, time, plan version	Traceability

All data must have defined units, time stamps and validation rules. Stale or missing values must not be treated as current.

4.6 External Interface Requirements

User interface: Web dashboard showing ship progress, crane and fleet status, yard occupancy, alerts and customs status, with role-based views.

Software interfaces: APIs or message streams to the terminal operating system, crane control systems, AGV fleet controller, customs systems, weather services and audit storage.

Hardware interfaces: Interfaces to crane controllers, AGVs/trucks, GPS/positioning sensors, RFID/OCR gate readers and yard cameras; servers with enough compute for real-time optimization.

Communication interface: Secure, authenticated, low-latency network (wired and wireless) with message validation, time-outs and redundancy.

5. Assumptions and Constraints

5.1 Assumptions

- Shipping lines supply accurate and timely cargo manifests and bay plans.
- Cranes, AGVs and yard equipment expose controllable and monitorable interfaces.
- Positioning and tracking sensors give accurate location data.
- Weather and arrival-time feeds are available and reasonably reliable.
- Customs systems provide hold and release status electronically.
- Operators will review and can override plans.
- The port has reliable network coverage across the terminal.
- Business rules and safety clearances are approved before deployment.

5.2 Constraints

Constraint Type	Description
Technical	Real-time optimization must run within strict time limits on large, changing data sets.
Physical	The yard footprint, rail layout and crane reach are fixed.
Operational	The system must run continuously and fit existing port workflows.
Safety	Personnel and equipment safety rules take priority over efficiency.
Economic	Costs of sensors, compute, integration and training are limited by budget.
Legal / Regulatory	Customs, maritime, data protection and port safety regulations apply.
Time-related	Design, integration and testing must fit the project schedule.
External dependency	Weather, vessel schedules and third-party systems are outside direct control.

6. Requirement Validation

6.1 Completeness and Consistency Checklist

Validation Item	Verification Method	Expected Result
Stakeholder needs	Map each stakeholder need to an FR or NFR	All major needs covered
Objective coverage	Check each of the five objectives against requirements	Every objective has related requirements
Functional completeness	Review data intake, planning, execution, adaptation and reporting	No workflow gap
Non-functional measurability	Define test and benchmark for each NFR	Each NFR has acceptance criteria
Ambiguity	Review terms such as “real time”, “quickly”, “optimal”	Replaced by measurable targets
Redundancy	Compare overlapping requirements	Each requirement has a distinct purpose
Consistency	Check conflicts between throughput and safety rules	Safety rules take priority
Traceability	Link requirements to use cases and test cases	Every requirement verifiable
Fault handling	Test sensor loss, crane failure and delayed vessels	Safe-response behaviour defined

6.2 Improvements Recommended

- Define “real time” precisely, with separate response limits for dashboards, re-planning and collision alerts.

- Define the optimization objective (for example vessel turnaround time, moves per hour, re-handling count) and how trade-offs are weighted.

- Specify a threshold for vessel delay that triggers automatic re-planning, and when manager approval is required.

- Add manual-override and fail-safe modes for crane and AGV control.
- Set availability, recovery time and data-loss targets.
- Define data retention and access rules for customs and audit data.
- Validate the models by simulation against historical port data before pilot use.

6.3 Requirement Traceability Matrix

Objective	Related Requirements	Verification
Balance yard placements	FR-05 to FR-07, NFR-12	Simulation of re-handling moves
Coordinate cranes, avoid collisions	FR-08 to FR-11, NFR-05	Conflict and fault simulation
Route trucks dynamically	FR-12, FR-13	Traffic and congestion tests
Adapt to ship delays	FR-14, FR-15	Delay scenario testing
Live dashboards for logistics and customs	FR-16 to FR-19, NFR-01, NFR-08	Usability and latency tests
Control, security and audit	FR-20 to FR-22, NFR-06, NFR-11	Security and audit review

7. Conclusion

This SRS defines the requirements for a Seaport Container Yard Coordination Platform that orchestrates container placement, crane schedules and terminal truck traffic. It identifies stakeholders, functional and non-functional requirements, actors, use cases, data, interfaces, assumptions and constraints. The platform aims to shorten ship turnaround, raise the utilization of cranes and automated trucks, reduce gridlock and fuel waste, and increase port capacity without expanding the physical footprint.

Before implementation, response-time targets, optimization objectives and safety fail-safes must be converted into measurable acceptance tests and validated through simulation and pilot trials.